

# Master Canonical Product System

The system is built around one core idea:

Every product sold across dispensary menus should connect back to a single organized product truth object.

That truth object is the **MCP — Master Canonical Product**.

Dispensaries may write product names differently, use inconsistent categories, misspell brands, shorten product lines, or leave out important details. The system's job is to normalize those messy menu listings and connect them to the correct product card.

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## Core Terms

### MCP — Master Canonical Product

The **Master Canonical Product** is the actual product card.

This is the clean, organized, system-wide product record. It represents the real product being sold, regardless of how different dispensaries title it on their menus.

Example:

#### **Stiiizy 1g Watermelon Z Liquid Diamonds Vape Pod**

This MCP can connect to many different dispensary listings for the same product.

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### PEK — Product Equality Key

The **Product Equality Key** is the fingerprint matching key.

It helps the system decide whether a scraped dispensary product listing belongs to an existing MCP or needs a new MCP created.

The PEK is not the product card itself. It is the matching logic behind the product card.

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## DPL — Dispensary Product Listing

The **Dispensary Product Listing** is the individual product as shown on a specific dispensary menu.

Each time a dispensary adds a product to its menu, the system pulls that item in as a DPL.

The DPL keeps the store-specific information, such as:

- Original menu title
- Price
- Sale price
- Stock status
- Purchase link
- Store image
- Raw platform data

The DPL connects back to the MCP.

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# Product Card Title Template

The MCP title should describe the exact sellable unit.

The title should follow this structure:

**Brand + Size + Strain/Product Line + Modifier + Product Form**

Or written another way:

**{Brand} {Size} {Line / Strain / Product Name} {Modifier} {Form}**

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## Title Components

### Brand

The company or product brand.

Example:

**Stiiizy**

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### Size / Weight / Count / Volume

The sellable unit size.

Examples:

- 1g
  - 0.5g
  - 3.5g
  - 7g
  - 10pk
  - 100mg
  - 5mg
  - 12oz
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## **Strain or Product Line**

The strain name, flavor name, or product line.

Examples:

- Watermelon Z
  - Blue Dream
  - Hash Burger
  - Apple Fritter
  - Pineapple Mango
  - Sour Diesel
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## **Extract / Infusion / Product Line Modifier**

The modifier describes the product style, infusion, extract type, or special product line.

Examples:

- Liquid Diamonds
  - Live Resin
  - Distillate
  - Rosin
  - Solventless
  - Hash Infused
  - THC
  - CBD
- 

## **Product Form**

The form is the specific shopper-facing product type.

Examples:

- Vape Pod
- Disposable
- Cartridge
- Infused Pre-Roll
- Beverage
- Gummy
- Flower
- Concentrate

- Rosin
  - Live Resin
  - Tincture
  - Capsule
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## Product Card Title Example

### Product Data

Brand:

**Stiiizy**

Size:

**1g**

Strain/Product Line:

**Watermelon Z**

Modifier:

**Liquid Diamonds**

Form:

**Vape Pod**

### Final MCP Title

**Stiiizy 1g Watermelon Z Liquid Diamonds Vape Pod**

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## More Product Title Examples

- Stiiizy 1g Watermelon Z Liquid Diamonds Vape Pod
- Stiiizy 1g Watermelon Z Liquid Diamonds Disposable
- MFNY 0.5g Hash Burger Live Resin Vape Cartridge
- Ayrloom 5mg Pineapple Mango THC Beverage

- Florist Farms 3.5g Blue Dream Flower
  - Jaunty 1g Sour Diesel Distillate Vape Cartridge
  - Heavy Hitters 2g Northern Lights All-in-One Disposable
  - Rove 1g Skywalker Live Resin Vape Cartridge
  - Veterans Choice 7g Apple Fritter Small Buds Flower
  - Off Hours 10pk Blue Raspberry Gummies
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## Category vs. Form

The system should separate **category** from **form**.

### Category

The category is the broad product class.

Examples:

- Vape
- Edible
- Flower
- Pre-Roll
- Concentrate
- Beverage
- Tincture
- Topical
- Accessory

### Form

The form is the exact product type.

Examples:

- Vape Pod
- Disposable
- Cartridge
- Gummy
- Chocolate
- Drink
- Infused Pre-Roll

- Rosin
- Resin
- Badder
- Sauce
- Flower Eighth

The visible MCP product card title should use the **form**.

The search filters should use both **category** and **form**.

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## Product Normalization Flow

Each scraped menu item becomes a **Dispensary Product Listing**.

The system then normalizes that listing, compares it against existing **Master Canonical Product** cards using the **PEK matching engine**, and decides whether to:

1. Attach the listing to an existing MCP
2. Send it to a match review queue
3. Create a provisional new MCP

The MCP becomes the core truth object that powers:

- Product search
  - Price comparison
  - Availability
  - Reviews
  - SEO pages
  - Brand pages
  - Category pages
  - Deal pages
  - Future signage rendering
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## MCP Data Structure

The MCP should hold a stable product identity.

These fields describe the actual product, not one store's version of the product.

mcp\_id  
pek  
canonical\_name  
display\_name  
brand\_id  
brand\_name  
category  
subcategory  
product\_type  
weight  
unit  
strain\_type  
canonical\_image\_url  
description  
known\_aliases  
known\_title\_variants  
first\_seen  
last\_seen  
confidence\_score  
review\_status

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# MCP Example

## Master Canonical Product

STIIIZY Blue Dream Vape Pod 1g

## Connected Dispensary Listings

- “Stiiizy Blue Dream 1G Pod” at Store A
- “STIIIZY - Blue Dream - Vape Pod - 1 gram” at Store B
- “Blue Dream Pod by Stiiizy 1g” at Store C

All three listings connect to one MCP because they represent the same exact sellable product.

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# Dispensary Product Listing Data Structure

Each store listing should hold volatile store-specific data.

This is the information that can change from one dispensary to another.

listing\_id  
mcp\_id  
store\_id  
platform  
vendor\_product\_id  
original\_title  
normalized\_title  
price  
sale\_price  
effective\_price  
in\_stock  
quantity\_available  
menu\_url  
purchase\_url  
image\_url\_from\_store  
raw\_category\_from\_store  
raw\_brand\_from\_store  
raw\_payload\_id  
last\_seen  
last\_checked

The DPL is not the product truth object. It is the store-level sighting of that product.

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## Matching Logic

The matching system should be layered.

Do not rely only on one PEK hash. That gets brittle fast.

The matching should use multiple signals:

1. Exact known platform product ID match
2. Exact alias match
3. Brand + normalized name + weight + category match

4. Fuzzy title similarity
  5. Image similarity or same CDN image
  6. Potency/package hints
  7. Manual review when confidence is not high enough
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## Match Confidence Rules

95–100 confidence = auto-attach to MCP  
80–94 confidence = attach but mark as needs review  
60–79 confidence = candidate match only  
Below 60 confidence = create provisional MCP

This prevents two different products from getting smashed into one card just because their names are kinda-sorta similar.

That is the raccoon-in-a-server-room failure mode.

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## Alias Learning

The MCP should absorb aliases over time.

Every time a listing is successfully attached to an MCP, the system should learn from that listing title.

Example aliases:

```
[  
  "Veterans Choice Blue Dream 1G Cart",  
  "Vet Choice Blue Dream Vape 1g",  
  "Veteran's Choice - Blue Dream Cartridge",  
  "Blue Dream 510 Cart by Veterans Choice"  
]
```

These aliases become future matching evidence.

This is how the engine gets smarter the longer it runs.

# Recommended Database Tables

## master\_products

This replaces or evolves the current `pek_master` idea into a stronger product-card table.

id  
pek  
canonical\_name  
display\_name  
brand\_id  
brand\_name  
category  
subcategory  
product\_type  
weight\_value  
weight\_unit  
strain\_type  
canonical\_image\_url  
description  
status  
confidence\_score  
created\_at  
updated\_at  
last\_seen

---

## product\_aliases

This table stores known title variants and alternate names.

id  
master\_product\_id  
alias\_text  
normalized\_alias  
source\_platform  
source\_store\_id  
confidence  
created\_at

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## dispensary\_product\_listings

This table stores each store-specific version of a product.

id  
master\_product\_id  
store\_id  
platform  
vendor\_product\_id  
original\_title  
normalized\_title  
price  
sale\_price  
effective\_price  
in\_stock  
quantity\_available  
menu\_url  
purchase\_url  
raw\_payload\_id  
first\_seen  
last\_seen  
last\_checked  
active

---

## product\_match\_candidates

This table stores uncertain matches for review.

id  
incoming\_listing\_id  
candidate\_master\_product\_id  
match\_score  
match\_reasons  
status  
reviewed\_by  
reviewed\_at

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## Suggested MCP Title Fields

The system should not store only the final product title.

It should store the parts used to build the title.

brand  
size\_value  
size\_unit  
strain\_or\_line  
product\_modifier  
form  
category  
canonical\_title  
search\_title

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## MCP Title Field Example

```
{  
  "brand": "Stiiizy",  
  "size_value": "1",  
  "size_unit": "g",  
  "strain_or_line": "Watermelon Z",  
  "product_modifier": "Liquid Diamonds",  
  "form": "Vape Pod",  
  "category": "Vape",  
  "canonical_title": "Stiiizy 1g Watermelon Z Liquid Diamonds Vape Pod",  
  "search_title": "stiiizy 1g watermelon z liquid diamonds vape pod"  
}
```

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## Matching Consequence

The title structure helps prevent dangerous false matches.

These should **not** collapse into one product card:

- Stiiizy 1g Watermelon Z Liquid Diamonds Vape Pod
- Stiiizy 1g Watermelon Z Liquid Diamonds Disposable
- Stiiizy 0.5g Watermelon Z Liquid Diamonds Vape Pod
- Stiiizy 1g Watermelon Z Live Resin Vape Pod

Same brand. Same strain. Different sellable product.

Different MCPs.

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## Final Rule

The canonical product card title should describe the exact sellable unit, not just the marketing name.

Clean formula:

**Brand + Size + Strain/Product Line + Modifier + Form**

That becomes the name of the truth object.

The messy dispensary titles become aliases underneath it.